Mohammadamin Rami

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EDUCATION

École Polytechnique, Télécom Paris, Institut Polytechnique de Paris — Paris, France
Master of Science in Machine Learning, Communications, and Security — Sept 2025 - Sept 2026
GPA: 17.38/20
Distinction: *Mention Très Bien (With Highest Honors)*

Sharif University of Technology — Tehran, Iran
Bachelor of Science in Electrical Engineering — Sept 2019 - July 2024
GPA: 18.17/20
Distinction: *With High Honors*

PUBLICATIONS AND PREPRINTS

NeurIPS Effective Context in Transformers: An Analysis of Fragmentation and Tokenization 2026
*A. Fesharaki, **M. Rami**, A. Tchamkerten*

Preprint Zero-Error Functional Source Coding with Limited-Lookahead Side Information 2026
*A. Fesharaki, **M. Rami**, A. Tchamkerten*

RESEARCH INTERESTS

- Theoretical Foundations of Machine Learning
- Information Theory
- High-Dimensional Statistics
- Mathematical Data Science

RESEARCH EXPERIENCE

Institute of Science & Technology Austria (ISTA) — Klosterneuburg, Austria
Scientific Researcher — Supervisor: Prof. Marco Mondelli — Nov 2026 – Present

- Interplay of generalization and model privacy and membership inference attack on over-parameterized high-dimensional models

Télécom Paris — Paris, France
Research Intern — Supervisor: Prof. Aslan Tchamkerten — March 2026 – Oct 2026

- Effective Context in Transformers: An Analysis of Fragmentation and Tokenization — theoretical analysis of how data representation fundamentally affects the achievable loss of finite-context next-token prediction models.
- Zero-Error Functional Source Coding with Limited-Lookahead Side Information — characterizing fundamental limits of zero-error functional source coding with causal and limited-lookahead side information at the decoder.

MCI (Mobile Communications of Iran) — Tehran, Iran
AI Research Intern — June 2024 – Sept 2024

- Down stream task fine-tuning of language encoders (BERT-based architectures) for Persian language

HONORS

- Ranked **1st** in MSc program cohort, Institut Polytechnique de Paris
- Ranked in the **top 15%** among Electrical Engineering undergraduates, Sharif University of Technology
- Ranked in the **top 0.05%** of participants in the Iranian National University Entrance Exam

RELEVANT COURSEWORK

Télécom Paris & École Polytechnique :

- *Information Theory* 18/20
- *Statistical Learning* 18.5/20
- *Deep Learning* 17.5/20
- *Convex Optimization* 17.3/20
- *Sequential Decision Making* 17.2/20
- *Advanced Coding* 18/20
- *Randomization in Computer Science* 15.6/20
- *Cryptography* 14/20
- *Multi-user Communications* 13.5/20

PROJECTS

Multimodal Sentiment Analysis *GitHub repository*

- Sentiment analysis of movie scenes by using the scene image and also the subtitle. We used CNNs for images, and LSTMs, Transformers, and BERT for text.

Fine Tuning RoBERTa *GitHub repository*

- Fine tuning RoBERTa for a sample down-stream task of sentiment analysis

Massive Traffic Data Analysis With PySpark *GitHub repository*

- Applying various big data algorithms(map-reduce, association rules, LSH, CUR, pixie, link analysis)on Tehran's traffic dataset to extract insight(most frequent origin- destinations, hourly traffic patterns, ...)

Computer Vision Playground *GitHub repository*

- This is a playground repository for training different computer vision architectures such as VGG-based and ResNet-based architectures and comparing their performance.

Numerical Implementation of Various Matrix Decompositions *GitHub repository*

- Implementing numeric algorithms for various matrix decomposition methods such as Schur Decomposition, SVD, Cholesky Decomposition, and Cholesky Decomposition.

Image De-noising Using Dictionary Learning

- De-noising images using combinations of various methods of sparse coding (MP, OMP, WMP), and dictionary learning (MOD, KSVD, A-KSVD)

SKILLS

Languages: Python, Matlab, C

Frameworks and Libraries: PyTorch, Transformers, TensorFlow, PySpark, Numpy, scikit-learn, Django, FastAPI, CVXPy, Pandas

Tools: Git/GitHub/GitLab, Kubernetes, Docker, SQL

Operating Systems: Linux

Document Creation: LaTeX, Markdown